

Pre-Greek substrate

The **Pre-Greek substrate** (or **Pre-Greek substratum**) consists of the unknown [pre-Indo-European](#) language(s) spoken in [prehistoric Greece](#) before the coming of the [Proto-Greek language](#) in the Greek peninsula during the [Bronze Age](#). It is possible that [Greek](#) acquired some thousand words and proper names from such a language or group of languages, because some of its vocabulary cannot be satisfactorily explained as deriving from Proto-Greek and a [Proto-Indo-European](#) reconstruction is almost impossible for such terms.^{[1][2]}

Introduction

Linguistic situation

Based upon toponymic and lexical evidence, it is generally assumed that one or several languages were once spoken in both the Greek peninsula and western [Asia Minor](#) before [Mycenaean Greek](#) and the attested [Anatolian languages](#) ([Hittite](#) and [Luwian](#)) became predominant in the region. Various explanations for this phenomenon have been given by scholars.^[3]

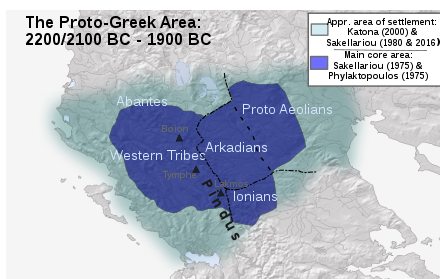
One [substrate language](#), whose influence is observable on Ancient Greek and Anatolian languages, is taken by a number of scholars to be an Indo-European language related to the Anatolian Luwian language,^{[4][5]} and to be responsible for the widespread place-names ending in -ssa and -nda in western Asia Minor, and -ssos and -nthos in mainland Greece, respectively.^{[6][7][4]} For instance, the name of the mount [Parnassos](#) in Greece has been interpreted as the Luwian *parna-* ('house') attached to the possessive suffix -ssa-. Both Hittite and Luwian texts also attest a place-name *Parnassa*, which could be related.^[4] Philologist [Martin L. West](#) has proposed to

name this unattested Anatolian language "Parnassian", and has argued for "a parallel movement down from [Thrace](#) by a branch of the same people as entered Anatolia, the people who were to appear 1,500 years later as the [Luwians](#)". From the distribution of the names, it appears that this language was spoken during the [Early Helladic II](#) period, which began around 2800 BC.^[4]

However, since many clusters of sounds are possible in both pre-Greek and Indo-European, it is difficult in most cases to distinguish between eventual "Parnassian" loanwords and shared pre-Indo-European substrate words. For instance, terms like *τολύπη* (*tolúpē*; 'clew, ball of wool ready for spinning') show typical pre-Greek features while being related to Anatolian words (in this case Luwian and Hittite *taluppa/i-* 'lump, clod') with no other attested Indo-European cognate, suggesting that they were borrowed from the same non-Indo-European source. Some scholars have thus proposed that at least part of the pre-Greek substrate was brought to Greece by [pre-Indo-European](#) settlers from Asia Minor, and that we should distinguish between different layers of loanwords coming successively or concurrently from different families of languages.^{[8][9][10]}

While the correlations between Anatolian and Greek placenames may be a strong indication of a common early phase of Indo-European – possibly Anatolian – influence in the area,^[11] some pre-Greek loanwords still remain incompatible with Indo-European phonology while showing certain recurrent patterns that set them apart from other languages.^[12] This likely indicates that "one language, or a group of closely related dialects or languages" was the source of another, possibly earlier, layer of pre-Indo-European loanwords in the region.^[13] Of the few words of secure Anatolian origin, most are cultural items or commodities which are likely the result of commercial exchange, not of a substratum.^[14] Some of the relevant vocabulary can also be explained as linguistic exchange between Greek and Anatolian languages across the [Aegean Sea](#) without necessarily originating from a change of language.^{[9][15]}

Coming of Proto-Greek



Proto-Greek area of settlement (2200/2100-1900 B.C.) suggested by Katona (2000), Sakellariou (2016, 1980, 1975) and Phylaktopoulos (1975)

Estimates for the introduction of the [Proto-Greek language](#) into prehistoric Greece have changed over the course of the 20th century. Since the decipherment of [Linear B](#), searches were made "for earlier breaks in the continuity of the material record that might represent the 'coming of the Greeks'".^[16]

The majority of scholars date the coming of Proto-Greek to the transition from [Early Helladic II](#) to Early Helladic III (c. 2400–2200/2100 BC),^[17] with the diversification into a southern and a northern group beginning by approximately 1700 BCE.^{[18][19][20][21][22][23]} This has been criticized by John E. Coleman, who argues that this estimate is based on stratigraphic discontinuities at [Lerna](#) that other archaeological excavations in Greece suggested were the product of chronological gaps or separate deposit-sequencing instead of cultural changes.^[24] Coleman estimates that the entry of Proto-Greek speakers into the Greek peninsula occurred during the late 4th millennium BC (c. 3200 BC) with pre-Greek spoken by the inhabitants of the Late Neolithic II period.^[25]

Reconstruction

Although no written texts exist or have been identified as pre-Greek, the [phonology](#) and [lexicon](#) have been partially reconstructed via the considerable number of words that have been borrowed into Greek; such words often show a type of variation not found in inherited Indo-European Greek terms, and certain recurrent patterns that can be used to identify pre-Greek elements.^[26]

Phonology

Vowels

The pre-Greek language had a simple [vowel system](#), with either three or five monophthongs. This system consisted of either /a/, /e/, /i/, /o/, /u/, or most likely just /a/, /i/, /u/, in which /a/ varied between /a/~e/~o/ as a result of palatalization for /e/ and labialization for /o/.

Additionally, it had at least one diphthong (/au/), and it may also have had /ou/, although this is also often explained as the sequence -ar^w- adapted in Greek as -oup-, since /ou/ is often seen with an /r/.^[27]

Consonants

The phonology of pre-Greek likely featured a series of both [labialized](#) and [palatalized](#) consonants, as indicated by [Mycenaean](#) inscriptions in [Linear B](#). These features were found not only in [stops](#), but in [resonants](#) as well (presumably including even the rare modified approximants /j^w/ and /w^j/), which was different from Indo-European languages at the time and is generally considered a rare feature characteristic of pre-Greek.^[27] It is, however, unlikely that [voicing](#) or [consonantal aspiration](#) were distinctive features, as pre-Greek loanwords in Greek vary freely between plain, voiced and aspirated stops (e.g. ἀσφάραγος/ἀσπάραγος, *aspháragos/aspáragos*, 'asparagus').^[27] The observation of such variants for a particular word is often a strong indication of substrate-derived etymology.

Furthermore, while the existence of word-initial [approximants](#) /w/ and /j/ can be safely inferred from common motifs in inherited words (e.g. the *ἰα-* from *ja- in ἱαμβος, ἰάσων) or even retained in early and dialectal forms (e.g. *wa- in the cases of ἀναξ-φάναξ, Ὀαξός-φαξός, ὑάκινθος-φάκινθος), [word-initial aspiration](#) probably did not exist, with /h/ considered by Beekes a non-native phoneme in pre-Greek.

The initial consonant σ- /s/ or /z/ is very common in pre-Greek and characteristic when it shows up as an s-mobile.

Consonant phonemes of Pre-Greek as posited by Beekes (2014)^[27]

	Labial			Dental, Alveolar			Palatal		Velar		
	plain	palatal	labial	plain	palatal	labial	plain	labial	plain	palatal	labial
Plosive	p	p ^j	p ^w	t	t ^j	t ^w			k	k ^j	k ^w
Nasal	m	(m ^j)	m ^w	n	n ^j	n ^w					
Fricative				s	s ^j	s ^w					
Trill				r	r ^j	r ^w					
Approximant				l	l ^j	l ^w	j	(j ^w)	w	(w ^j)	

Consonant clusters

Certain characteristic [consonant clusters](#) associated with pre-Greek phonology as reflected in words inherited into Greek, as listed by Beekes according to their frequency in the PIE language:^[27]

Not possible in PIE

- -κχ- /kk^h/, not possible in PIE, only in pre-Greek (but rare)
- -πφ- /pp^h/, not possible in PIE, though still very rare in pre-Greek
- -τθ- /tt^h/, impossible in PIE, common in pre-Greek

Rare in PIE

- -βδ- /bd/, rare in PIE, not as much in pre-Greek
- -γδ- /gd/, rare in PIE, not in pre-Greek
- -δν- /dn/, rare in PIE, not in pre-Greek
- -ρκν- /rkn/, very rare overall and found only in pre-Greek loans
- -σβ- /sb/, very rare and problematic identification in PIE, common in pre-Greek probably from *-sg^w-
- -σγ- /sg/, rare in PIE, common in pre-Greek perhaps from *-tīg-
- non word-initial -σκ- /sk/ and -στ- /st/, rare in PIE, somehow common in pre-Greek derivative words
- -χμ- /k^hm/ and -χν- /k^hn/, rare in PIE, sometimes in substrate words
- word-initial ψ- /ps/, extremely common in pre-Greek loans (most words beginning with ψ- being such)

Possible in PIE

- -γν- /gn/, not as rare in both PIE and pre-Greek
- -κτ- /kt/, common in PIE but in pre-Greek also with variants -χθ-, -χτ- etc.
- -μν- /mn/, common in PIE and also in many pre-Greek words
- -ρδ- /rd/, possible in PIE, also found in some pre-Greek words
- -ρν- /rn/, when pre-Greek usually also with variants -ρδ- and -νδ-
- -στλ- /stl/, possible in PIE but more common in substrate words
- -φθ- /p^ht^h/, possible in PIE but also common in pre-Greek loans

Lexicon

There are different categories of words that have been suggested to be pre-Greek (or "Aegean") loanwords such as:^{[28][29]}

- **Anatomy:**

- αὐχὴν, *aukhēn*, 'neck';
- λαιμός, *laimós*, 'neck, throat';
- ῥίς, *rhís*, 'nose, snout';
- σιαγών, *siagōn*, 'jaw, jawbone';
- σπόνδυλος/σφόνδυλος, *spōndylos/sphōndylos*, 'vertebra';
- σφάκελος/σφάκηλος, *sphákelos/sphákēlos*, 'middle finger'.

- **Animals:**

- ἀράχνη, *arákhnē*, 'spider';
- βόλινθος/βόνασσοι, *bólinthos/bónassos*, 'wild ox';
- κάνθαρος, *kántharos*, 'beetle';
- κῆτος, *kētos*, 'whale, sea monster';
- πελεκῖνος, *pelekīnos*, 'pelican';
- σμίνθος, *smínthos*, 'mouse'.

- **Architecture and building materials:**

- ἄργιλλος/ἄργιλος/ἄργιλλα, *árgillos/árgilos/árgilla*, 'white clay, argil';
- καλύβη/καλυβός/κολυβός, *kalybē/kalybós/kolybós*, 'hut, cabin';
- λαβύρινθος, *labýrinthos*, 'labyrinth';
- πέτρα, *pétrā*, 'stone (as building material)';
- πλίνθος, *plínthos*, 'brick';
- πύργος, *pýrgos*, 'tower'.^[30]

- **Geography and topography:**

- ἄμβων/ἄμβη, *ámboŋ/ámbē*, 'crest of a hill', 'raised edge or protuberance';^[10]
- κρημνός, *krēmnoś*, 'edge of a trench, cliff';
- κορυφή, *koruphé*, 'mountain top';

- ὄχθη, *ókhthē*, 'riverbank';
- σπέος/σπεῖος, *spéos/speíos*, 'cave, cavern';
- χαράδρα/χαράδρη, *kharádra/kharádrē*, 'torrent, riverbed, gorge';
- **Maritime vocabulary:**
 - ἄκατος, *ákatos*, 'small dinghy, skiff'.
 - θάλασσα, *thálassa*, 'sea'.
 - θάλαμος, *thálamos*, 'an inner room or chamber', 'the lowest, darkest part of the ship', 'the hold';^[10]
 - θίς, *thís*, 'heap of sand, beach, shore, sand at the bottom of the sea';
 - κυβερνάω, *kybernáo*, 'to steer, to be a steerman'.
- **Metals and metallurgy:**
 - κασσίτερος, *kassíteros*, 'tin';
 - μόλυβδος, *mólybdos*, 'lead';
 - σίδηρος, *sídēros*, 'iron';
 - τάγχουρος/τάγχαρας/ἄγχουρος, *tánkhouros/tánkharas/ánkhouros*, 'gold';
 - χαλκός, *chalkós*, 'copper'.
- **Musical instruments:**
 - κίθαρις, *kítharis*, 'zither';
 - λύρα, *lýra*, 'lyre';
 - σάλπιγξ, *sálpinx*, 'trumpet';
 - σύριγξ, *sýrinx*, 'flute';
 - φόρμιγξ, *phórminx*, 'lyre'.
- **Mythological characters:**
 - Ἀχιλλεύς/Ἀχιλεός, *Akhilleús/Akhileús*, *Achilles*;^[31]
 - Δαναός, *Danaós*, *Danaus*;
 - Κάδμος, *Kádmos*, *Cadmus*;
 - Ὀδυσσεύς, *Odysseús*, *Odysseus*;^[32]

- Ῥαδάμανθους, *Rhadámanthys*, [Rhadamanthus](#).

- **Plants:**

- ἄμπελος, *ámpelos*, 'vine';
- ἀψίνθιον, *apsínthion*, 'wormwood' or '[absinthe](#)';
- ἐλαία, *elaía*, 'olive tree';
- κισσός, *kissós*, 'ivy';
- κολοκύνθη/κολοκύνθα/κολοκύνθος/κολοκύντη, *kolokýnthē/kolokýnthā/kolokýnthos/kolokýntē*, 'bottle gourd';
- κυπάρισσος, [kypárisos](#), 'cypress';
- σταφυλή, *staphylḗ*, 'grape';
- σῦκον/τῦκον, *sýkon/týkon*, 'fig'.

- **Social practices and institutions:**

- ἄμιλλα, *hámilla*, 'contest, trial, sporting activity';
- ἀτέμβω, *atémbō*, 'maltreat' or 'to be bereft or cheated of a thing';^[10]
- ῥάναξ/ἄναξ, *wánax/ánax*, 'lord, king';
- θίασος, *thíasos*, '[thiasus](#), Bacchic revel';
- τύραννος, *týrannos*, 'absolute ruler'.

- **Theonyms:**

- Ἀπόλλων, *Apóllōn*, [Apollo](#);^{[33][34]}
- Ἄρης, *Arēs*, [Ares](#);^[34]
- Ἄρτεμις, *Ártemis*, [Artemis](#);^[34]
- Ἀσκληπιός, *Asklēpiós*, [Asclepius](#);^[35]
- Ἀθήνη, *Athénē*, [Athena](#);^[34]
- Ἄτλας, *Átlās*, [Atlas](#);^[35]
- Διόνῡσος, *Dióh̄ysos*, [Dionysus](#);^[35]
- Ἑρμῆς, *Hermēs*, [Hermes](#);^[36]
- Ἥφαιστος, *Héphaistos*, [Hephaestus](#);^{[37][36]}

- Ἰαπετός, *Īapetós*, [Iapetus](#).^[36]

- **Tools related to agricultural activities:**

- δίκηλλα, *díkella*, 'adze, pickaxe';
- κάμαξ, *kámax*, 'vine pole';
- μάκελλα, *mákella*, 'mattock, pick';
- χαλινός, *khalīnós*, 'bridle, rein'.

- **Toponyms/placenames:**

- -νθ-, -nth- (e.g. Κόρινθος, *Kórinthos*, [Corinth](#); Ζάκυνθος, *Zákynthos*, [Zakynthos](#));
- -σσ-, -ss- (e.g. Παρνασσός, *Parnassós*, [Parnassus](#));
- -ττ-, -tt- (e.g. Ἀττική, *Attiké*, [Attica](#); Ὑμηττός, *Hymēttós*, [Hymettus](#));^[38]
- region names e.g. Ἀχαΐα, *Achaíā*, [Achaea](#); Λακωνία, *Lakōnía*, [Laconia](#); Μαγνησία, *Magnēsíā*, [Magnesia](#);
- city names e.g. Δωδώνη, *Dōdōnē*, [Dodona](#); Κνωσσός, *Knōssós*, [Knossos](#); Κυδωνία, *Kydōnía*, [Cydonia](#);
- isles e.g. Κρήτη, *Krētē*, [Crete](#); Νάξος, *Náxos*, [Naxos](#);
- mountain names e.g. Ὀλυμπος, *Ólympos*, [Olympus](#); Πίνδος, *Píndos*, [Pindus](#);
- [hydronyms](#) e.g. Ἀχελῷος, *Akhelôios*, [Achelous](#); Γέλας, *Gélās*, [Gela](#); Ἴλισός, *Īlīsós*, [Ilisos](#);
- other geographical features e.g. Σούνιον, *Sóúnion*, [Sounion](#);
- mythological locations e.g. Ἠλύσιον, *Ēlýsion*, [Elysium](#).

- **Use of domestic species:**

- ἔλαιον, *élaion*, 'olive oil';
- λήκυθος, *lékythos*, 'oil-flask';
- κάνθων, *kánthōn*, 'pack-ass';
- στέμφυλον, *stémphylon*, 'mass of olives from which the oil has been pressed, mass of pressed grapes'.

- **Weapons:**

- θώραξ, *thórax*, 'corselet';
- μάστιξ, *mástīx*, 'whip';

- ὕσσός, *hyssós*, 'javelin'.
- **Weaving:**
 - ἀρύβαλλος, *arýballos*, 'purse';
 - βρόχος, *brókhos*, 'slip knot, mesh';
 - ἡλακάτη, *ēlakátē*, 'spindle';
 - μύρινθος, *mýrinthos*, 'cord'.

Anatolian loanwords

Possible Anatolian or "Parnassian" loanwords include:^[15]

- Ἀπόλλων, *Apóllōn* (Doric: *Apéllōn*, Cypriot: *Apeílōn*), from **Apeljōn*, as in [Hittite](#) *Appaliunaš*,^[33]
- δέπας, *dépas* 'cup; pot, vessel', Mycenaean *di-pa*, from [Hieroglyphic Luwian](#) *ti-pa-s* 'sky; bowl, cup' (cf. Hittite *nēpis* 'sky; cup');
- ἐλέφας, *eléphās* 'ivory', from Hittite *lahpa* (itself from Mesopotamia; cf. [Phoenician](#) 'lp, [Egyptian](#) *ḳbw*);
- κύανος, *kýanos* 'dark blue glaze; enamel', from Hittite *kuwannan-* 'copper ore; azurite' (ultimately from [Sumerian](#) *kù-an*);
- κύμβαχος, *kýmbachos* 'helmet', from Hittite *kupaḫi* 'headgear';
- κύμβαλον, *kýmbalon* 'cymbal', from Hittite *ḫuḫupal* 'wooden percussion instrument';
- μόλυβδος, *mólybdos* 'lead', Mycenaean *mo-ri-wo-do*, from **mork^w-io-* 'dark', as in [Lydian](#) *mariwda(ś)-k* 'the dark ones';
- ὄβρυζα, *óbryza* 'vessel for refining gold', from Hittite *ḫuprušḫi* 'vessel';
- τολύπη, *tolýpē* 'ball of wool', from Hittite *taluppa* 'lump'/'clod' (or [Cuneiform Luwian](#) *taluppa/i*).

Other substratum theories

Other explanations have been made for these [substrate](#) features. Some [fringe theories](#) ranging from the mild (e.g., [Egyptian](#)) to the extreme (e.g., [Proto-Turkic](#)) have been proposed but have been given little to no consideration from the broader academic community and as such are not developed in this article.

Minoan substratum

The existence of a [Minoan](#) ([Eteocretan](#)) substratum was the opinion of [English](#) archaeologist [Arthur Evans](#) who assumed widespread Minoan colonisation of the Aegean, policed by a Minoan [thalassocracy](#).^[39]

Raymond A. Brown, after listing a number of words of pre-Greek origin from [Crete](#), suggests a relation between [Minoan](#), [Eteocretan](#), [Lemnian](#) ([Pelasgian](#)), and [Tyrrhenian](#), inventing the name "Aegeo-Asianic" for the proposed language family.^[40]

However, many Minoan loanwords found in Mycenaean Greek (e.g., words for architecture, metals and metallurgy, music, use of domestic species, social institutions, weapons, weaving) have been asserted to be the result of socio-cultural and economic interactions between the Minoans and Mycenaeans during the Bronze Age, and may therefore be part of a linguistic [adstrate](#) in Greek rather than a substrate.^[41]

Tyrrhenian substratum

A [Tyrrhenian](#)/Etruscan substratum was proposed on the basis of the [Lemnos funerary stele](#):^[42] four pottery sherds inscribed in Etruscan that were found in 1885 at Ephestia in [Lemnos](#).^[42]

However, the Lemnos funerary stele was written in a form of [ancient Etruscan](#), which suggested that the author had emigrated from Etruria in [Italy](#), rather than the Greek sphere, and the [Homeric tradition](#) makes no mention of a Tyrrhenian presence on Lemnos.^[43]

If Etruscan was spoken in Greece, it must have been effectively a [language isolate](#), with no significant relationship to or interaction with speakers of pre-Greek or ancient Greek, since, in the words of C. De Simone, there are no Etruscan words that can be "etymologically traced back to a single, common ancestral form with a Greek equivalent".^[43]

Kartvelian theory

In 1979, Edzard J. Furnée proposed a theory by which a pre-Greek substrate is associated with the [Kartvelian languages](#).^[44]

See also

- [Camunic language](#) (probably Raetic)
- [Elymian language](#) (probably Indo-European)
- [Eteocretan](#)
- [Eteocypriot](#)
- [Etruscan language](#)
- [Hattic language](#)
- [Hurro-Urartian languages](#)
 - [Hurrian language](#)
 - [Urartian language](#)
 - [?Kassite language](#)
- [Minoan language](#)
- [North Picene language](#)
- [Paleo-Sardinian language](#) (also called Paleosardinian, Protosardic, [Nuraghic language](#))
- [Proto-Greek language](#)
- [Sicanian language](#)
- [Sicel language](#)
- [Tyrrhenian languages](#)

Substrates of other Indo-European languages

- [Germanic substrate hypothesis](#)
- [Goidelic substrate hypothesis](#)
- [Old European hydronymy](#)
- [Substratum in Vedic Sanskrit](#)

Citations

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1. [Duhoux 2007a](#), pp. 220–222.

2. [Beekes 2014](#), pp. 47–48: "Our knowledge of Indo-European has expanded so much, especially in the last thirty years (notably because of the laryngeal theory) that in some cases we can say almost with certainty that an Indo-European reconstruction is impossible. [...] In my EDG, I marked with >PG< all words which, in my view, were of Pre-Greek origin. I found 1106 words."
3. [Furnée 1972](#), p. 35; [Renfrew 1998](#), pp. 253–254; [Finkelberg 2006](#), p. 52; [West 2007](#), p. 8; [Beekes 2009](#), p. 3
4. [West 2007](#), p. 8.
5. Some scholars, such as [Leonard R. Palmer](#), go so far as to suggest that the language of [Linear A](#) might be Luwian, though other Anatolian interpretations have also been offered.
6. [Renfrew 1998](#), pp. 253–254.
7. [Finkelberg 2006](#), p. 52: "As we have seen, the suffixes -nth- and -ss-, which a hundred years ago gave rise to the hypothesis of the non-Indo-European pre-Hellenic substratum, can now be accounted for as typically Anatolian or, to be more precise, Luwian."
8. [Furnée 1972](#), p. 35.
9. [Beekes 2009](#), p. 3.
10. [Mihaylova 2012](#), pp. 80–81.
11. [Renfrew 1998](#), pp. 253–254, 256–257.
12. [García-Ramón 2004](#), pp. 999–1000.
13. [Beekes 2014](#), pp. 1, 45.
14. [Beekes 2009](#), p. xv.
15. [Hajnal 2009](#), pp. 1–21.
16. [Coleman 2000](#), p. 104.
17. [Meier-Brügger 2017](#), p. 697; citing Strunk 85–98, Panagl 99–103, and Lindner 105–108 in [Bammesberger & Vennemann 2003](#).
18. Hajnal, Ivo (2007). "Die Vorgeschichte der griechischen Dialekte: ein methodischer Rück- und Ausblick". In Hajnal, Ivo; Stefan, Barbara (eds.). [Die altgriechischen Dialekte. Wesen und Werden. Akten des Kolloquiums, Freie Universität Berlin, 19.–22. September 2001](#) (<https://www.academia.edu/2382692>) (in German). Innsbruck, Austria: Institut für Sprachen und Literaturen der Universität Innsbruck. p. 136.
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20. Woodard, Roger D. (2008). *The Ancient Languages of Europe* (<https://books.google.com/books?id=aPEE NAEp938C&q=north%20versus%20south&pg=PA52>) . Cambridge University Press. p. 52. ISBN 978-1-139-46932-6.
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24. Coleman 2000, pp. 106–107.
25. Coleman 2000, p. 139ff.
26. Beekes 2014, p. 1.
27. Beekes 2014, p. 6–10.
28. Renfrew 1998, pp. 244–245 (see Tables 1 and 2 for all loanwords except personal names, toponyms and theonyms).
29. Beekes 2014.
30. If the substratum is actually Indo-European, *pyrgos* as well as *Pergamos* might be connected to Proto-Indo-European **bhergh-* (<http://www.bartleby.com/61/roots/IE59.html>) Archived (<https://web.archive.org/web/20081015064110/http://www.bartleby.com/61/roots/IE59.html>) 2008-10-15 at the Wayback Machine.
31. Beekes 2009, p. xvii and 183.
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33. Beekes 2003, pp. 1–21.
34. Beekes 2014, p. 160.
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38. [Renfrew 1998](#), pp. 241, 253–254.
39. [Gere 2006](#), p. 112: "Arthur Evans would live to repent of his suggestion to the British School that they reopen the excavations at Mycenae. He had expected that his theory of Minoan dominance over the mainland would be borne out, but instead he encountered stout resistance... Evans could never bring himself to believe any story except that of Minoan colonisation of the mainland from the beginning to the end of Mycenaean history."
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